Cell Design Specification — VBF-T7R1NOFORCE-DS-01

Case t7_r1_noforce (HEV battery) — 2026-08-25. PDF release of design_spec.md; values from parameter set / tool outputs.

	Value	Source
	NMC811 (positive) / graphite (negative), Chen2020 baseline	parameter set
acity	5.0 Ah / 5.0065 Ah	param set / r5_slim_1c_c
midpoint	2.5–4.2 V / 3.826 V	param set / calc-energy
	65 x 1580 mm (unfolded); stack 186.8 um; volume 19.18 cm3	param set / calc-energy
	LiPF6/carbonate (Chen2020); design transport: sigma 2.5 S/m, D 5e-10 m2/s, t+ 0.5 (bridge estimates)	params_r5_slim.json
	75.6 um NMC811, porosity 0.335, rho 3262 kg/m3 (unchanged)	Chen2020
e	85.2 um graphite, porosity 0.25, rho 1657 kg/m3, particles 4.0 um, Al2O3 ALD coating (SEI k 7.5e-13 m/s)	params_r5_slim.json
	10 um (eps 0.47, rho 397) / Al 10 um, Cu 6 um	params_r5_slim.json
	0.667 (spec formula: neg 56.74 Ah/m2 / pos 85.03 Ah/m2); usable-window 1.00 by charge balance	calc.xlsx sheet 4
	36.226 g (contract caliber, electrolyte excluded); 42.55 g incl. electrolyte est.	calc-energy / sec.3
/ neg / sep	163.99 / 105.88 / 2.10 g/m2	calc-energy layer_kg_m2
neg	2.169 / 1.243 g/cm3 (x(1-eps) / 1000)	formula
	5.27 cm3 pore volume -> 6.33 g @1.2 g/cm3 (lit), fill factor 1.0 assumed	sec.3
	0.1C CC to 4.2 V, 25 C, 2 cycles (design recommended; production needs tuning)	sec.3
	491.47 Wh/kg (928.06 Wh/L) vs >=327.18 -> PASS	r5_slim_energy.json
/ criterion	anode min +0.0153 V -> plated=false -> PASS	r5_slim_4c45.json
C / criterion	510.51 nm vs <=550 -> PASS (+39.5 nm)	r5_slim_aging45.json
on	triggered=false (T_final 317.0 K) -> PASS	r5_slim_nail.json
sity	2.6415 mOhm / 43.24 kW/kg	calc-energy